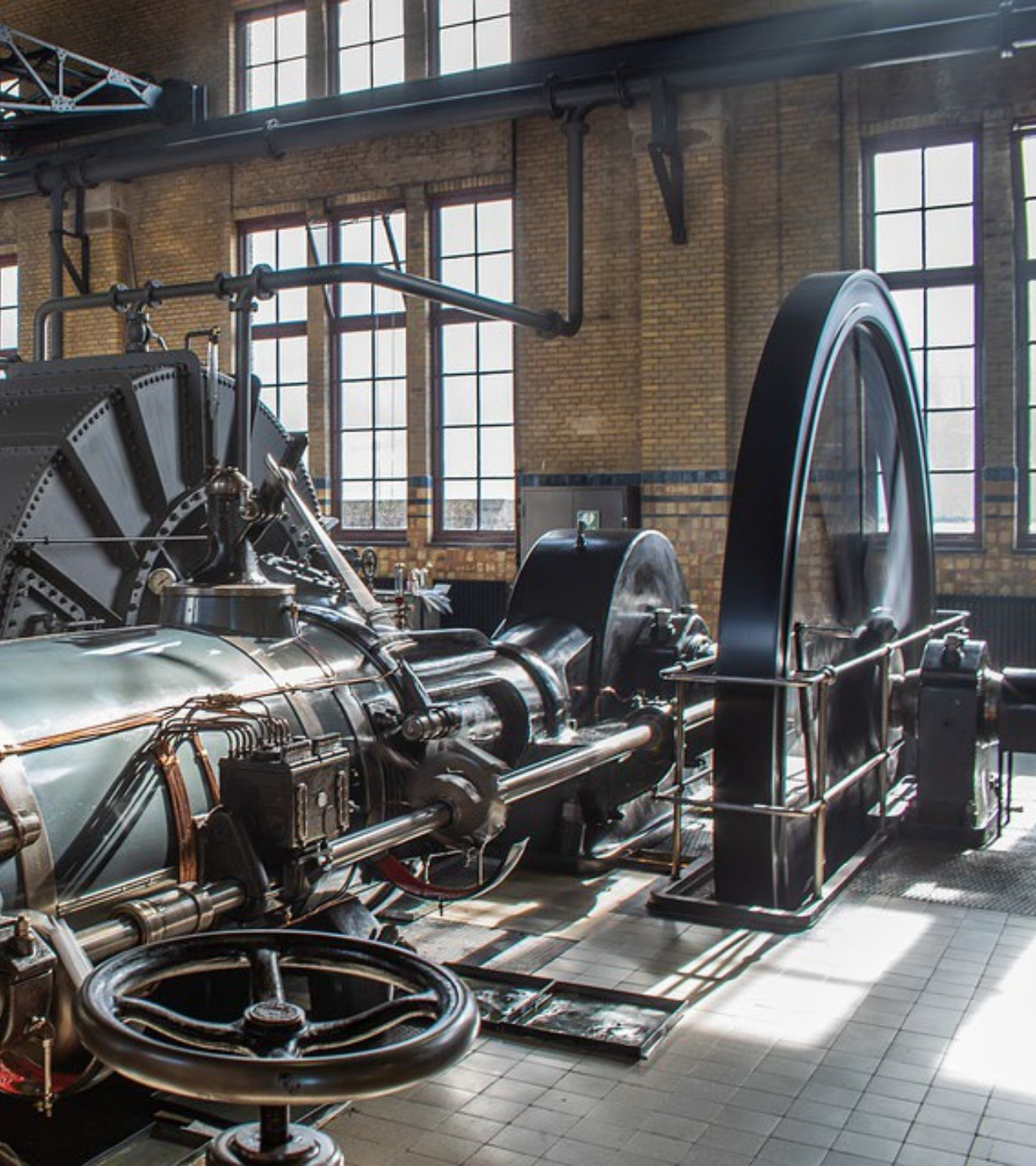
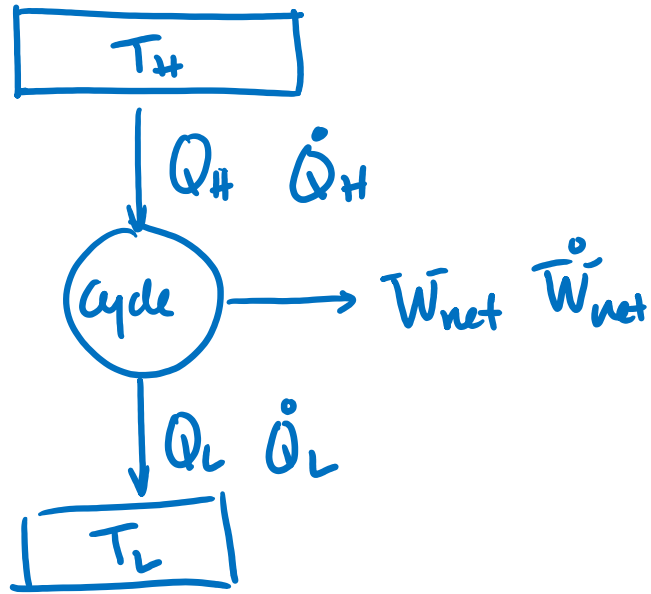




ME 300:
Engineering Thermodynamics

Heat Engines and Cycle Analysis

Heat engine



$$\left. \begin{aligned} \bar{W}_{net} &= \bar{W}_{out} - \bar{W}_{in} \\ Q_H &= Q_{in} \\ Q_L &= Q_{out} \end{aligned} \right\}$$

$$Q_H = \bar{W}_{net} + Q_L$$

$$\eta_{th} = \frac{\bar{W}_{net}}{Q_H}$$

$$\eta_{carnot} = 1 - \frac{T_L}{T_H} \quad \text{max, cycle is reversible}$$

Heat engine – example

Energy from a high-temperature reservoir at 750 K flows at a rate of 2.1 kW to a heat engine. If the engine rejects heat to a low-temperature reservoir at 300 K, what is the maximum possible power the engine can deliver? What is the rate of heat rejection?

$$T_H = 750 \text{ K}$$

$$\dot{Q}_H = 2.1 \text{ kW}$$

$$T_L = 300 \text{ K}$$

$$\begin{aligned} \text{max power} \rightarrow \eta_{th} &= \eta_{\text{carnot}} = 1 - \frac{T_L}{T_H} \\ &= 1 - \frac{300}{750} \rightarrow \eta_{th, \text{max}} = 0.6 \end{aligned}$$

Assume: steady flow

$$\begin{aligned} \eta_{th} &= \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}} \rightarrow \dot{W}_{\text{net}, \text{max}} = \eta_{th, \text{max}} \dot{Q}_{\text{in}} \\ &= (0.6)(2.1) \end{aligned}$$

$$\dot{W}_{\text{net}, \text{max}} = 1.26 \text{ kW}$$

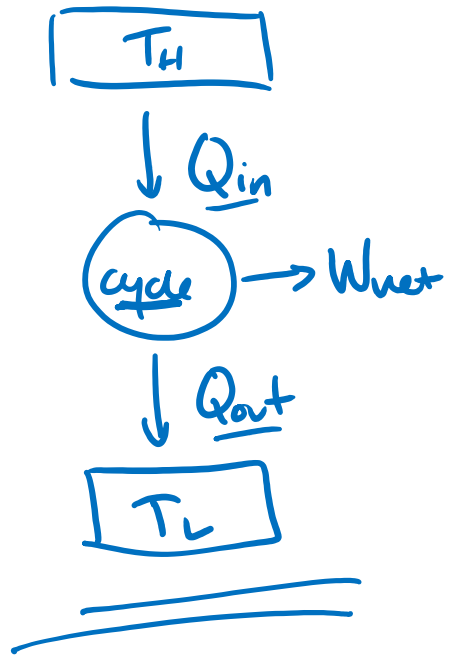


What is the rejected heat in this problem?

$$\dot{Q}_H^\vee = \dot{W}_{net}^\vee + \dot{Q}_L$$

$$\begin{aligned}\dot{Q}_L &= \dot{Q}_H - \dot{W}_{net} \\ &= 2.1 - 1.26 \\ &= \underline{0.84 \text{ kW}}\end{aligned}$$

Cycle analysis



Power cycles: extract power from working fluid (WF)

Process 1-2: raise energy of WF by work $\Rightarrow \bar{W}_{in}$

Process 2-3: raise energy of WF by heat $\Rightarrow Q_{in}$

Process 3-4: extract energy via work $\Rightarrow \bar{W}_{out}$

Process 4-1: reject heat $\Rightarrow Q_{out}$

$$\bar{W}_{net} = \bar{W}_{in} + \bar{W}_{out} > 0$$
$$< 0 \quad > 0$$

Examples of cycles

Stirling cycle : $\Delta T = 0$ comp.
: $\Delta V = 0$ heat in
: $\Delta T = 0$ exp.
: $\Delta V = 0$ heat out

Carnot cycle : $\Delta S = 0$ comp
: $\Delta T = 0$ heat in
: $\Delta S = 0$ exp.
: $\Delta T = 0$ heat out

Rankine (steam) : $\Delta S = 0$ comp.
 $\Delta P = 0$ heat in
 $\Delta S = 0$ exp
 $\Delta P = 0$ heat rej.

Air Standard

Brayton : $\Delta S = 0$ comp
: $\Delta P = 0$ heat in
: $\Delta S = 0$ exp.
: $\Delta P = 0$ heat out

Diesel : $\Delta S = 0$ comp
: $\Delta P =$ heat in
: $\Delta S = 0$ exp
: $\Delta V = 0$ heat out

Otto : $\Delta S = 0$ comp
 $\Delta V = 0$ heat in
 $\Delta S = 0$ exp
 $\Delta V = 0$ heat out